Lab-4 (1) Static and Default Routes(E)

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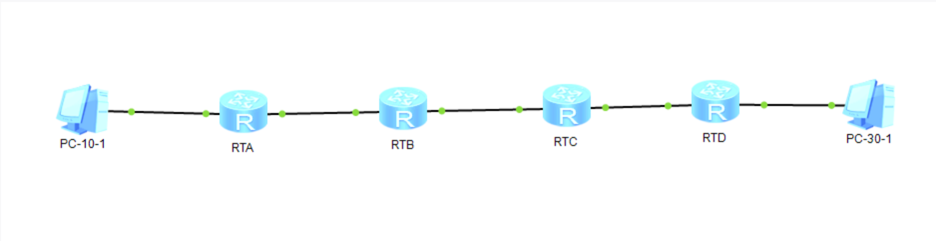
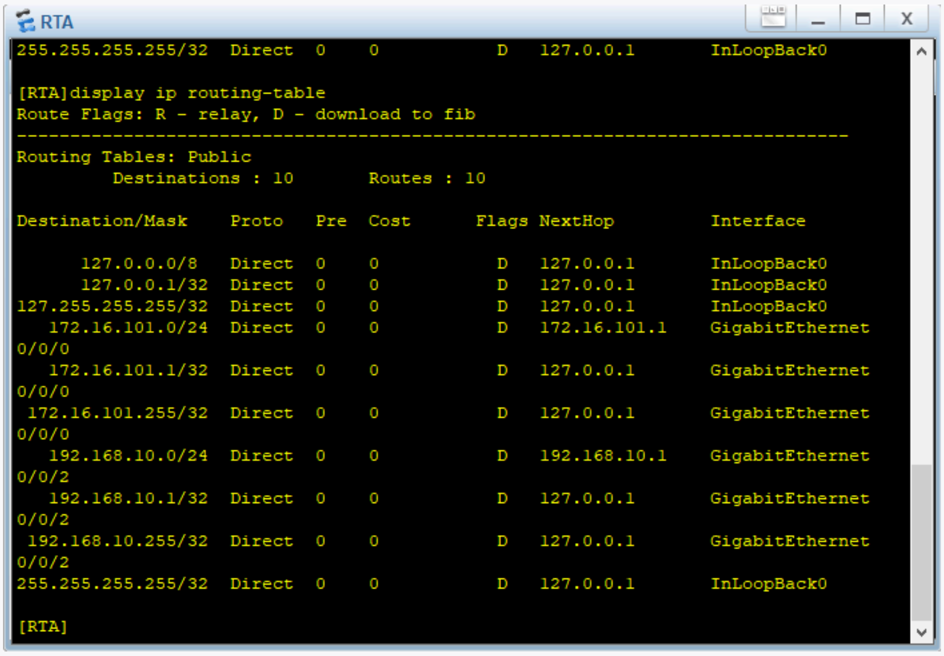
Lab-4-1 STATIC AND DEFAULT ROUTE CONFIGURATION

Q4-1.1. Paste the screenshot of the created topology.

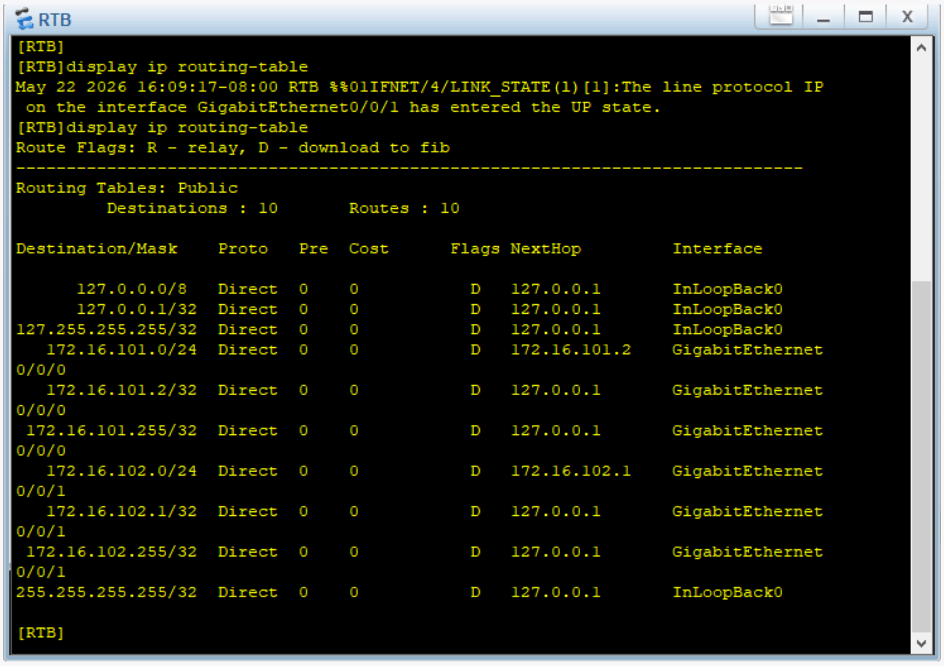


Q4-1.2. Paste the screenshot of the IP routing table on the router RTA.



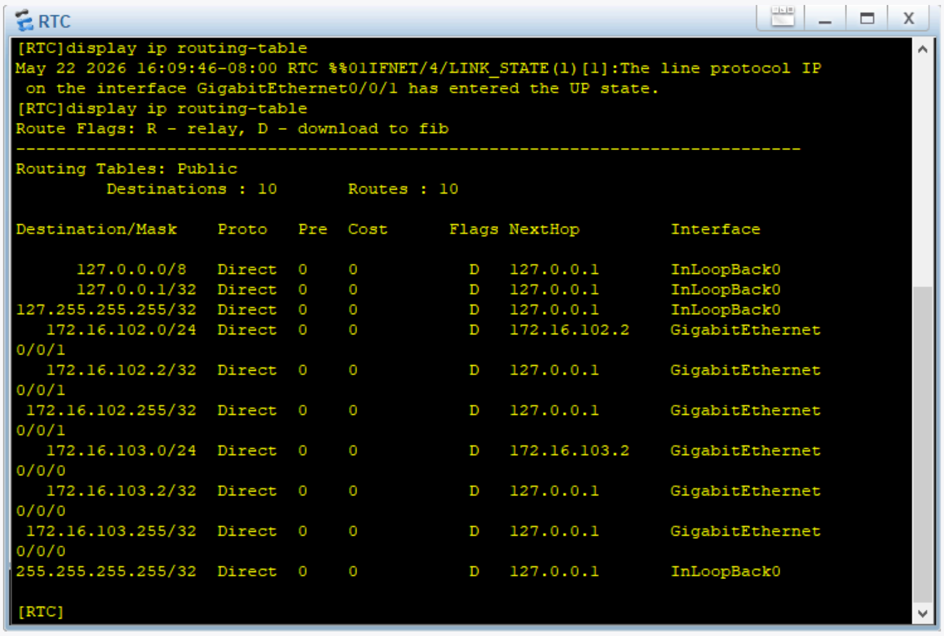

```
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTA]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 10       Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.101.0/24  Direct  0    0           D   172.16.101.1    GigabitEthernet
0/0/0
   172.16.101.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.101.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.10.0/24  Direct  0    0           D   192.168.10.1    GigabitEthernet
0/0/2
   192.168.10.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
 192.168.10.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTA]
```

Q4-1.3. Paste the screenshot of the IP routing table on the router RTB.



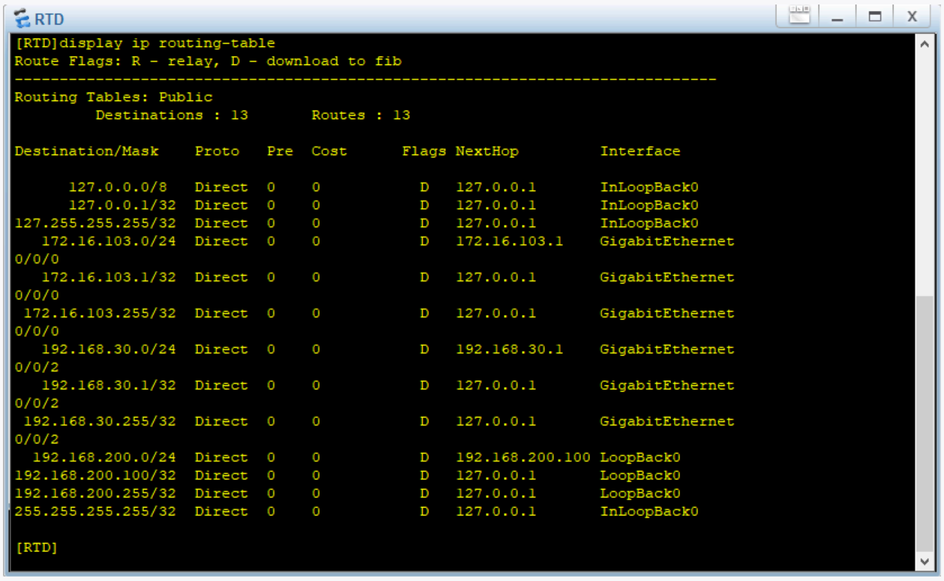
```
RTB
[RTB]
[RTB]display ip routing-table
May 22 2026 16:09:17-08:00 RTB %%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
 on the interface GigabitEthernet0/0/1 has entered the UP state.
[RTB]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 10       Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.101.0/24  Direct  0    0           D   172.16.101.2    GigabitEthernet
0/0/0
   172.16.101.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.101.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   172.16.102.0/24  Direct  0    0           D   172.16.102.1    GigabitEthernet
0/0/1
   172.16.102.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
 172.16.102.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTB]
```

Q4-1.4. Paste the screenshot of the IP routing table on the router RTC.



```
RTC
[RTC]display ip routing-table
May 22 2026 16:09:46-08:00 RTC %%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
 on the interface GigabitEthernet0/0/1 has entered the UP state.
[RTC]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 10       Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.102.0/24  Direct  0    0           D   172.16.102.2    GigabitEthernet
0/0/1
   172.16.102.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
 172.16.102.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
   172.16.103.0/24  Direct  0    0           D   172.16.103.2    GigabitEthernet
0/0/0
   172.16.103.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.103.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTC]
```

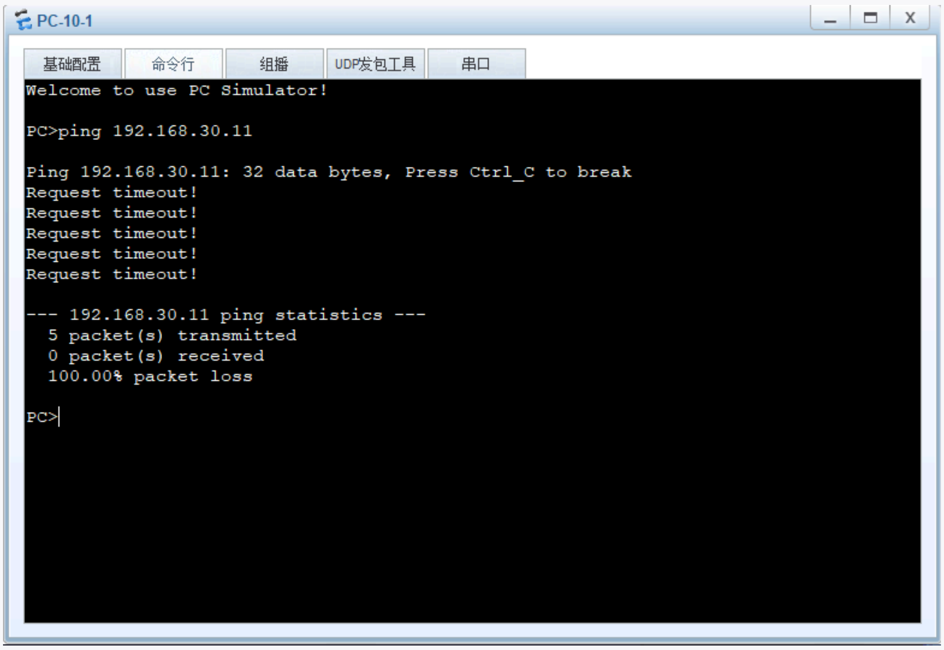
Q4-1.5. Paste the screenshot of the IP routing table on the router RTD.



```
RTD
[RTD]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 13       Routes : 13

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.103.0/24  Direct  0    0           D   172.16.103.1    GigabitEthernet
0/0/0
   172.16.103.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.103.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.30.0/24  Direct  0    0           D   192.168.30.1    GigabitEthernet
0/0/2
   192.168.30.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
 192.168.30.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
  192.168.200.0/24  Direct  0    0           D   192.168.200.100 LoopBack0
192.168.200.100/32  Direct  0    0           D   127.0.0.1       LoopBack0
192.168.200.255/32  Direct  0    0           D   127.0.0.1       LoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTD]
```

Q4-1.6. Can PC-10-1 and PC-30-1 ping each other? Please paste the screenshot of the ping result.

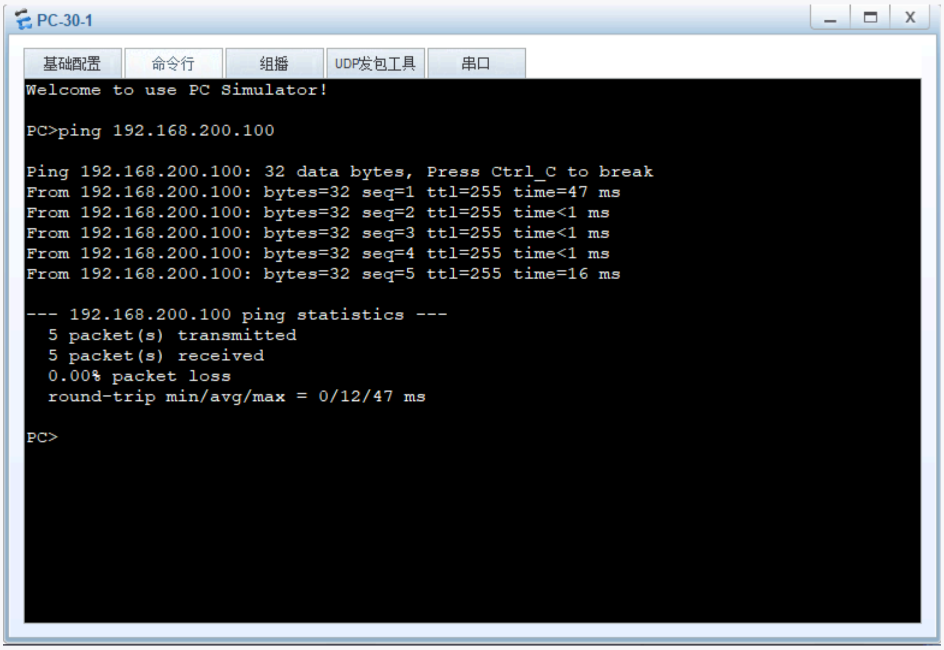
不能。



```
PC-10-1
基础配置  命令行  组播  UDP发包工具  串口
Welcome to use PC Simulator!

PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 192.168.30.11 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
  100.00% packet loss

PC>
```

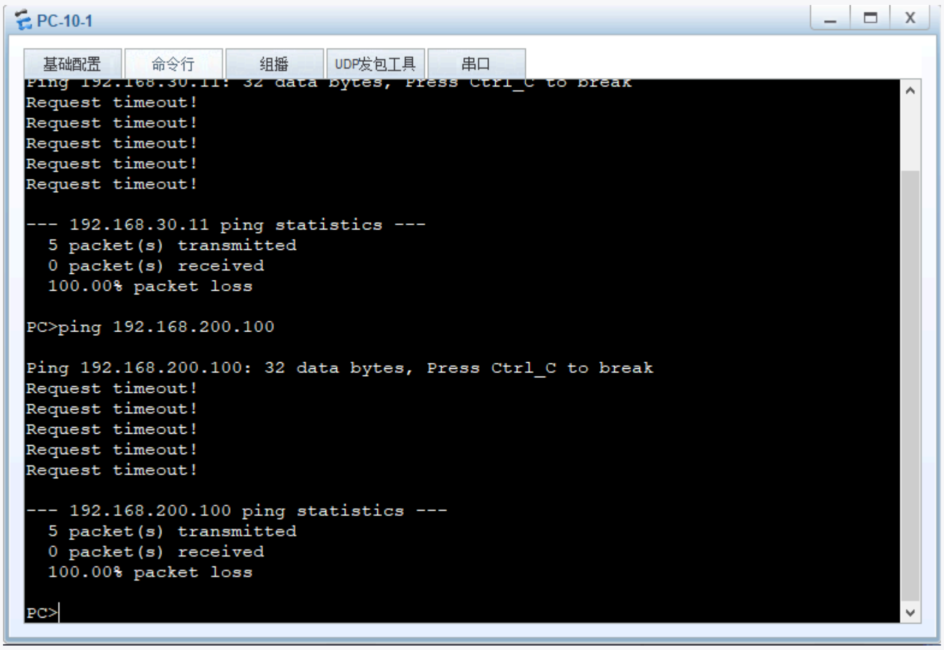
Q4-1.7. Can PC-30-1 and loopback 0 on RTD ping each other? Please paste the screenshot of the ping result.

可以。

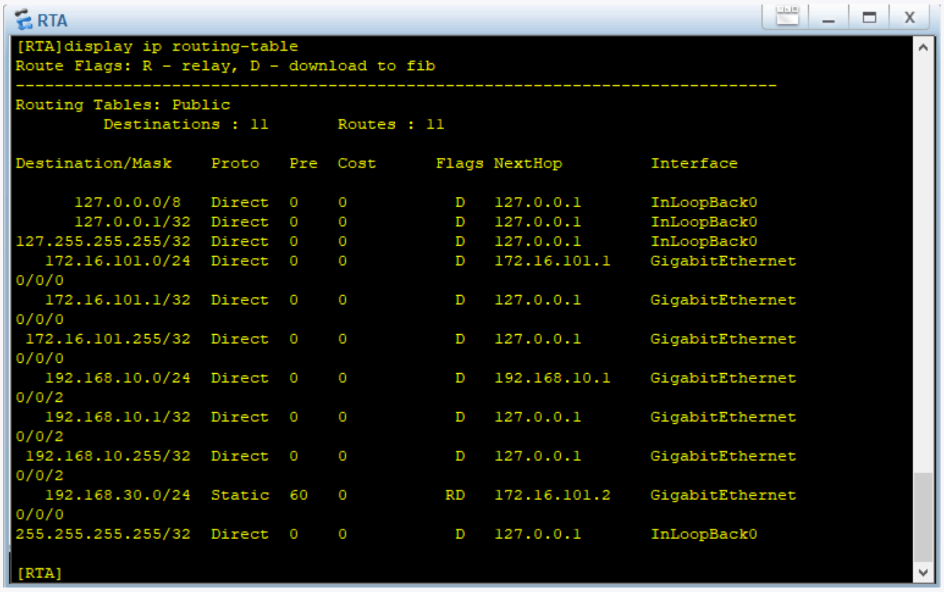


Q4-1.8. Can PC-10-1 and loopback 0 on RTD ping each other? Please paste the screenshot of the ping result.

不能。

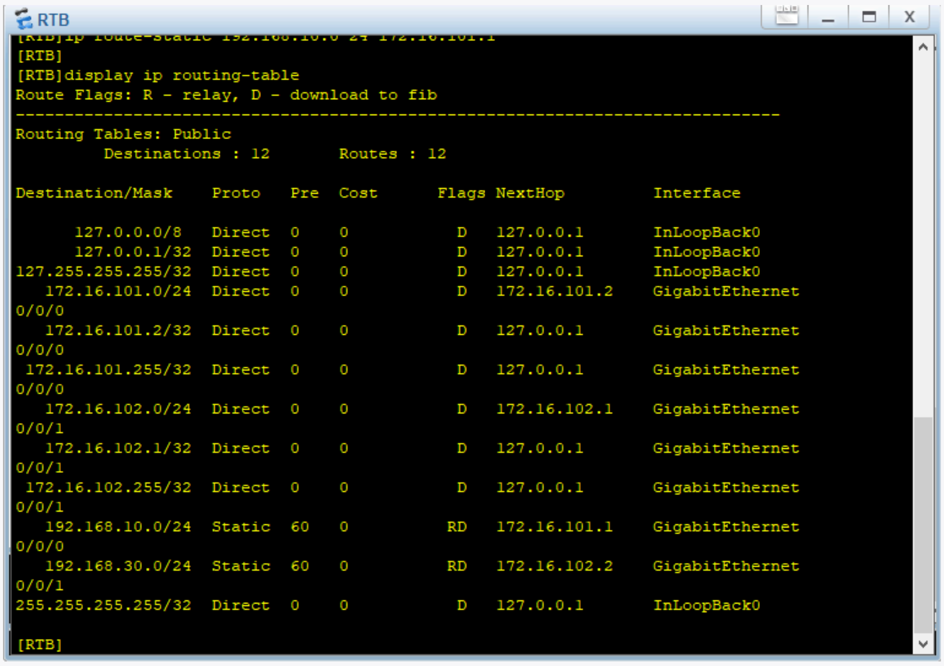


Q4-1.9. Paste the screenshot of the IP routing table on the router RTA.



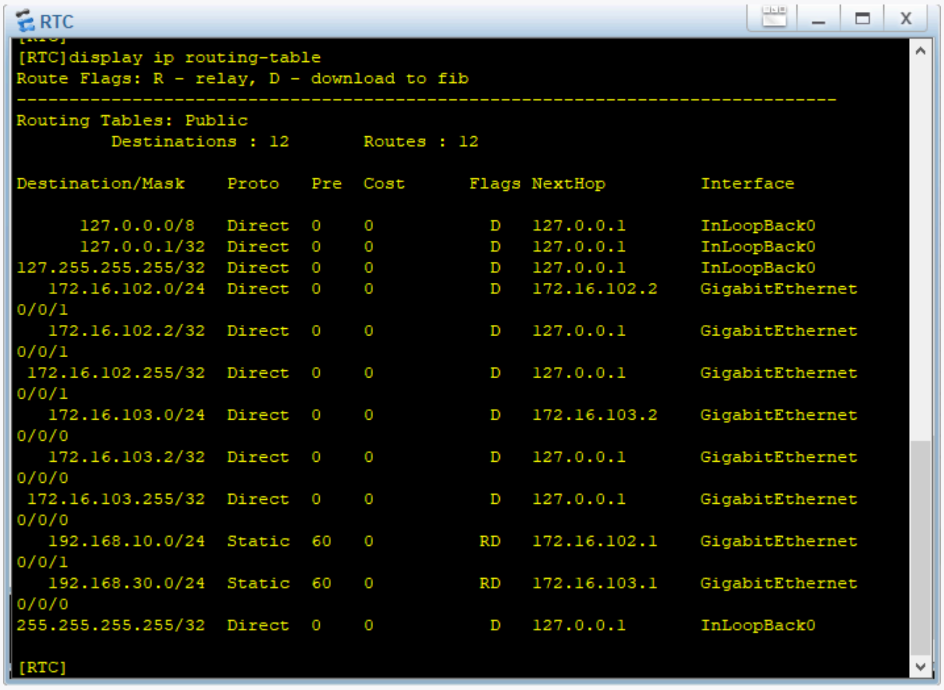
```
[RTA]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 11       Routes : 11

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.101.0/24  Direct  0    0           D   172.16.101.1    GigabitEthernet
0/0/0
   172.16.101.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.101.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.10.0/24  Direct  0    0           D   192.168.10.1    GigabitEthernet
0/0/2
   192.168.10.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
 192.168.10.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
   192.168.30.0/24  Static  60   0          RD   172.16.101.2    GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTA]
```

Q4-1.10. Paste the screenshot of the IP routing table on the router RTB.



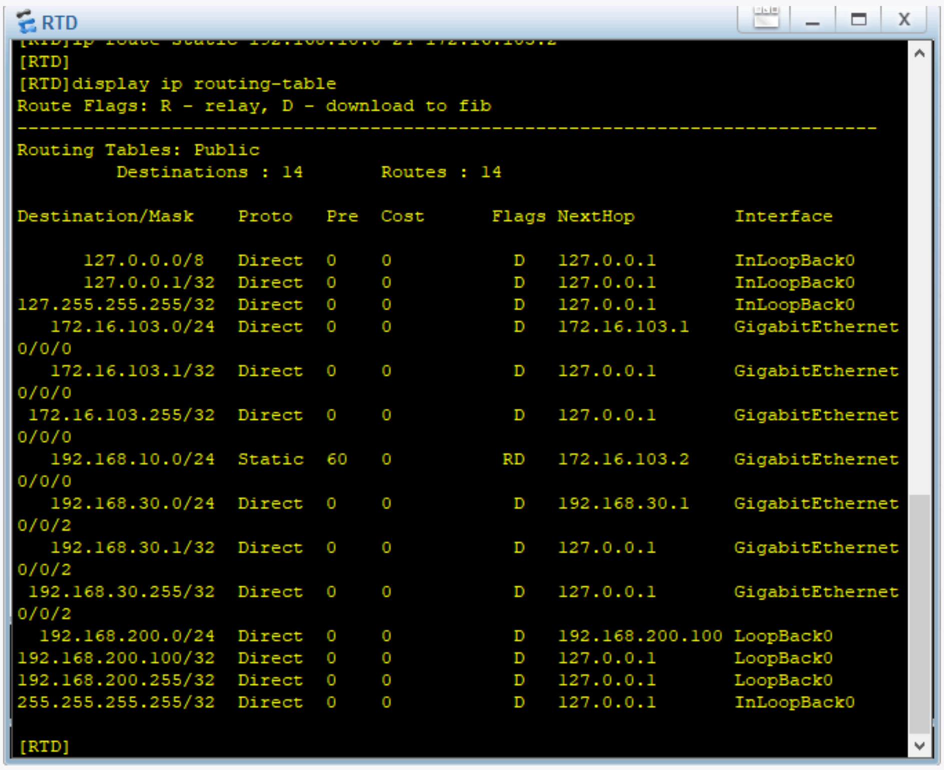
```
[RTB]
[RTB]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 12       Routes : 12

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.101.0/24  Direct  0    0           D   172.16.101.2    GigabitEthernet
0/0/0
   172.16.101.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.101.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   172.16.102.0/24  Direct  0    0           D   172.16.102.1    GigabitEthernet
0/0/1
   172.16.102.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
 172.16.102.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
   192.168.10.0/24  Static  60   0          RD   172.16.101.1    GigabitEthernet
0/0/0
   192.168.30.0/24  Static  60   0          RD   172.16.102.2    GigabitEthernet
0/0/1
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTB]
```

Q4-1.11. Paste the screenshot of the IP routing table on the router RTC.



```
RTC
[RTC]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 12       Routes : 12

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.102.0/24  Direct  0    0           D   172.16.102.2    GigabitEthernet
0/0/1
   172.16.102.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
 172.16.102.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
   172.16.103.0/24  Direct  0    0           D   172.16.103.2    GigabitEthernet
0/0/0
   172.16.103.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.103.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.10.0/24  Static  60   0          RD   172.16.102.1    GigabitEthernet
0/0/1
   192.168.30.0/24  Static  60   0          RD   172.16.103.1    GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTC]
```

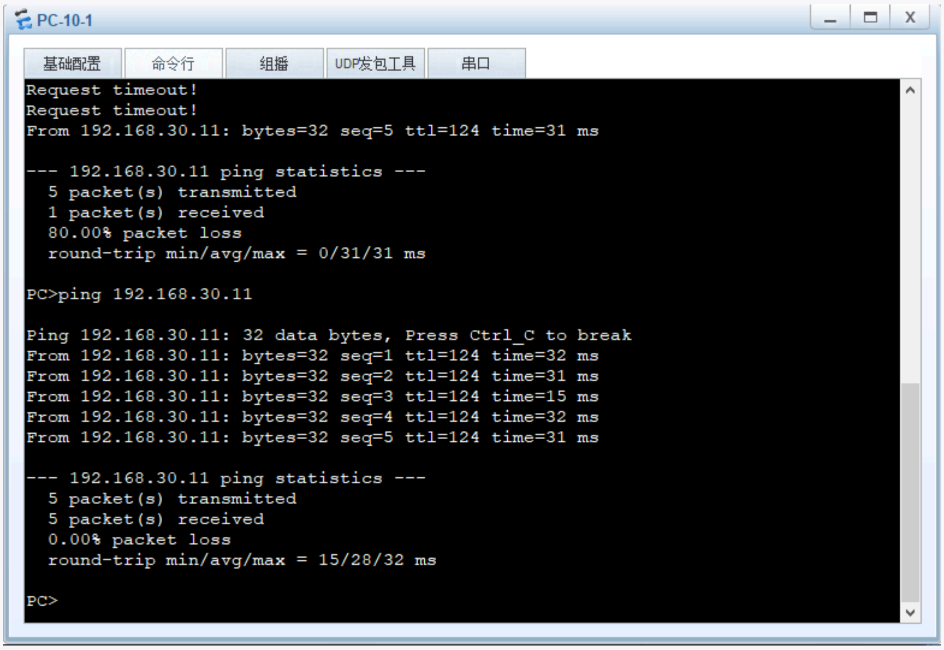
Q4-1.12. Paste the screenshot of the IP routing table on the router RTD.



```
RTD
[RTD]
[RTD]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 14       Routes : 14

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.103.0/24  Direct  0    0           D   172.16.103.1    GigabitEthernet
0/0/0
   172.16.103.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.103.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.10.0/24  Static  60   0          RD   172.16.103.2    GigabitEthernet
0/0/0
   192.168.30.0/24  Direct  0    0           D   192.168.30.1    GigabitEthernet
0/0/2
   192.168.30.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
 192.168.30.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
  192.168.200.0/24  Direct  0    0           D   192.168.200.100 LoopBack0
192.168.200.100/32  Direct  0    0           D   127.0.0.1       LoopBack0
192.168.200.255/32  Direct  0    0           D   127.0.0.1       LoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTD]
```

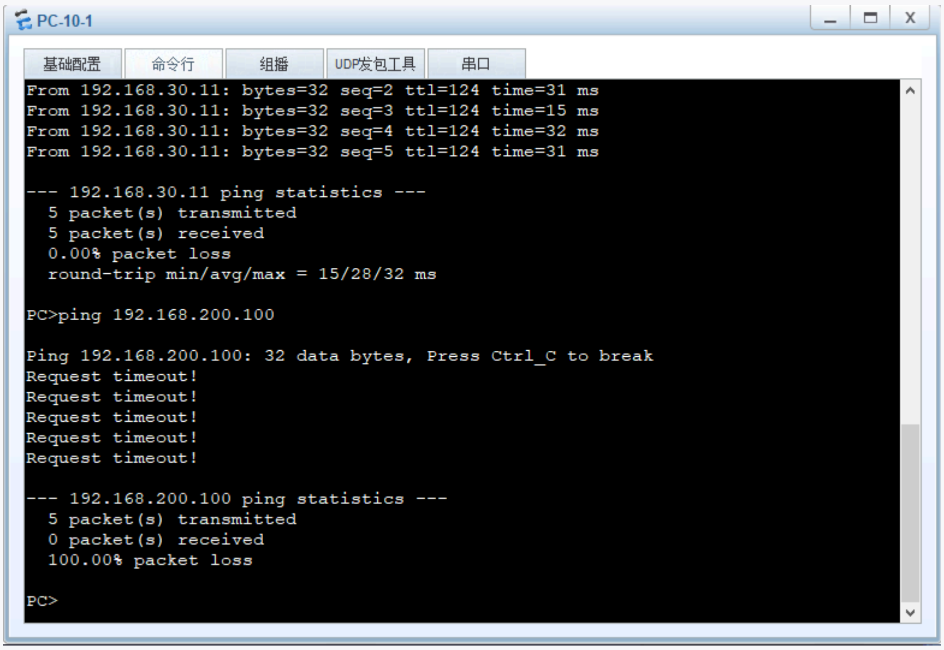
Q4-1.13. Can PC-10-1 and PC-30-1 ping each other? Please paste the screenshot of the ping result.

可以。

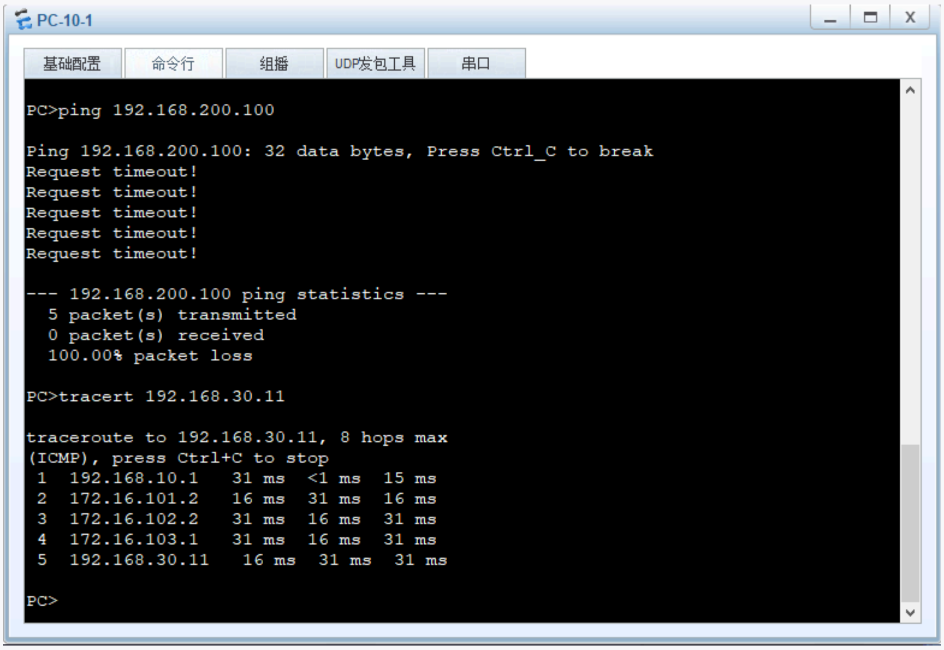


Q4-1.14. Can PC-10-1 and loopback 0 on RTD ping each other? Please paste the screenshot of the ping result. If they cannot ping each other, please explain why.

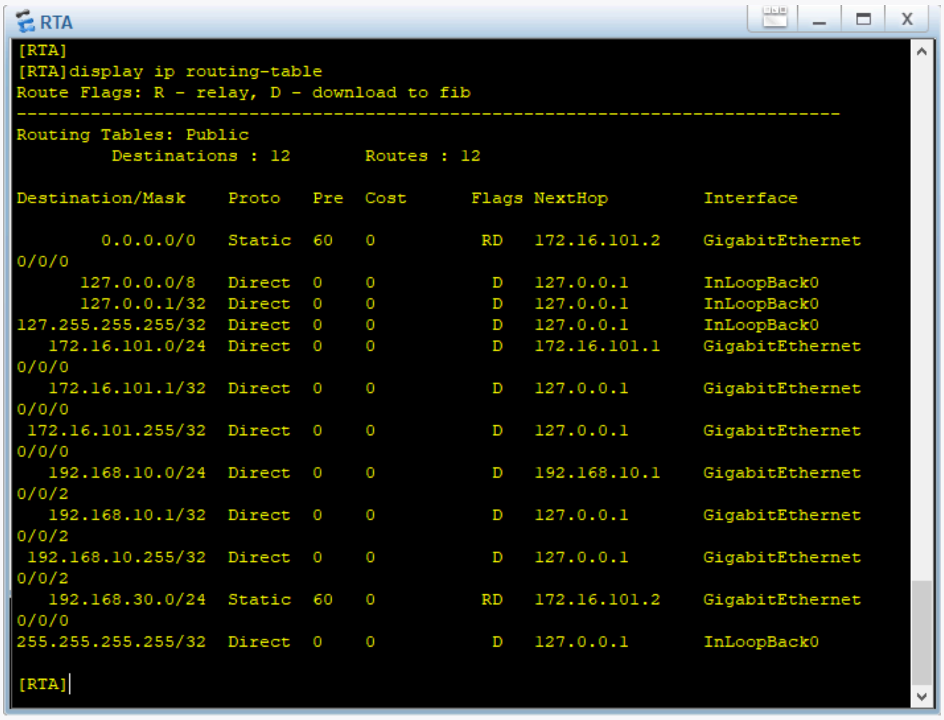
不能。因为各路由器（RTA、RTB、RTC）的路由表中均未配置到达 RTD 的 Loopback 0（192.168.200.0/24）网段的明细路由，导致数据包无法找到下一跳转发路径，在传输中途被丢弃。



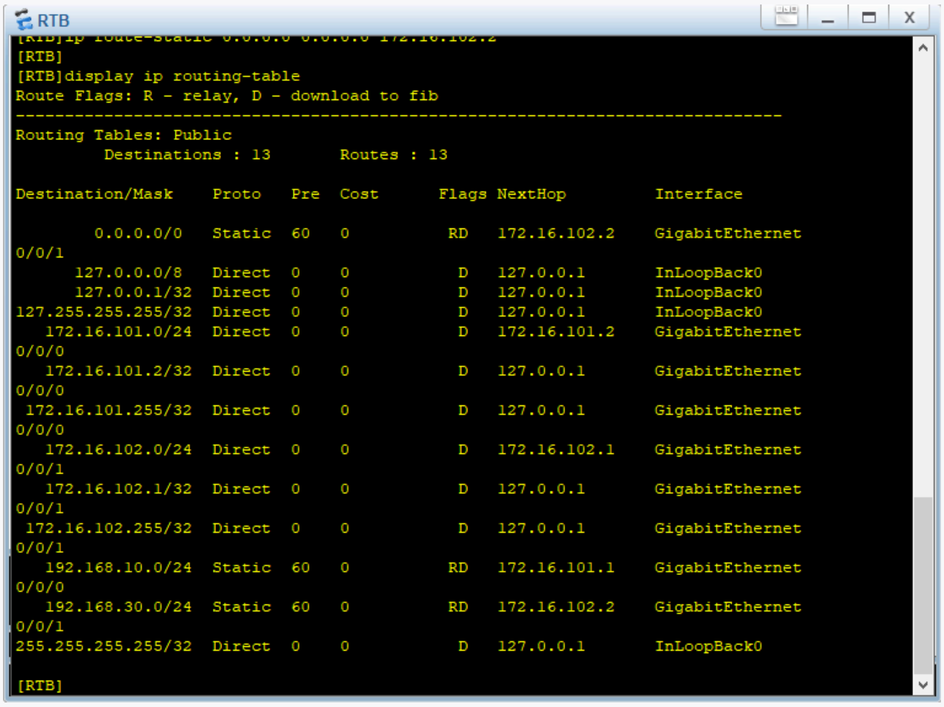
Q4-1.15. Please paste the screenshot of the result of command "tracert 192.168.30.11" issued from PC-10-1.



Q4-1.16. Paste the screenshot of the IP routing table on the router RTA.

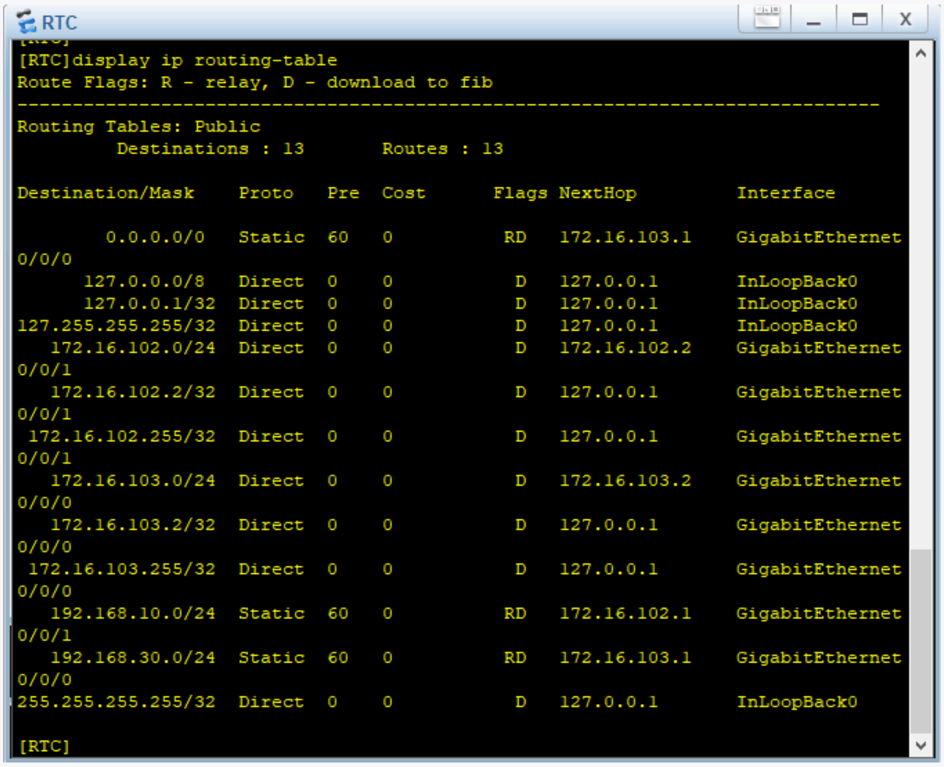


Q4-1.17. Paste the screenshot of the IP routing table on the router RTB.



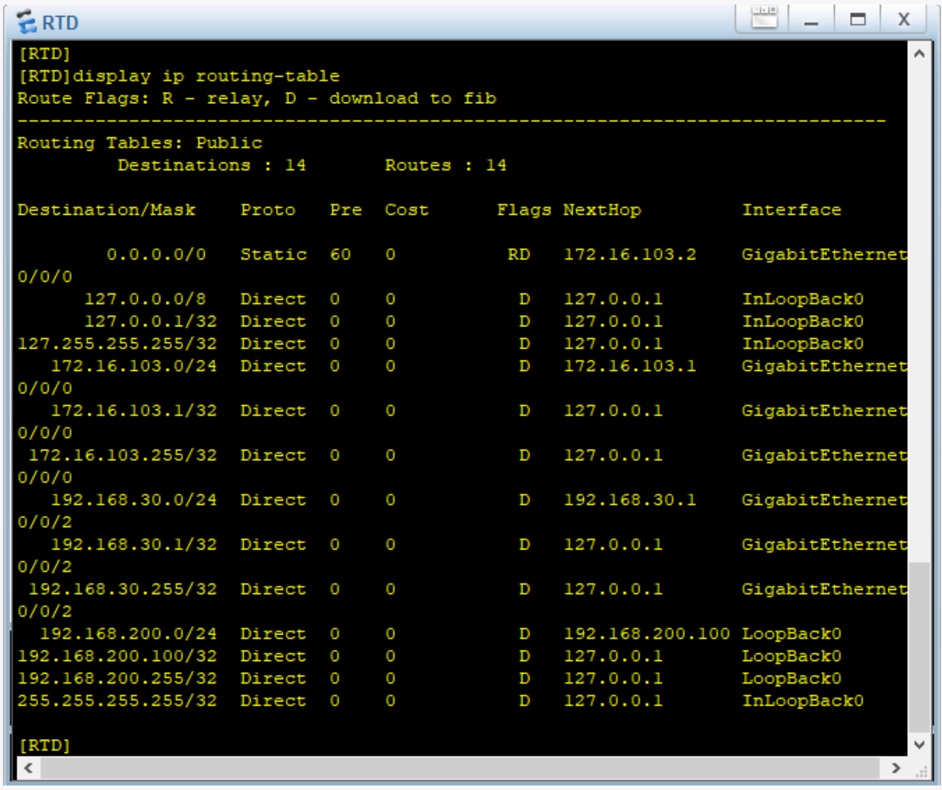
```
RTB
[RTB]
[RTB]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 13       Routes : 13

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0          RD   172.16.102.2    GigabitEthernet
0/0/1
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.101.0/24  Direct  0    0           D   172.16.101.2    GigabitEthernet
0/0/0
   172.16.101.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.101.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   172.16.102.0/24  Direct  0    0           D   172.16.102.1    GigabitEthernet
0/0/1
   172.16.102.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
 172.16.102.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
   192.168.10.0/24  Static  60   0          RD   172.16.101.1    GigabitEthernet
0/0/0
   192.168.30.0/24  Static  60   0          RD   172.16.102.2    GigabitEthernet
0/0/1
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTB]
```

Q4-1.18. Paste the screenshot of the IP routing table on the router RTC.



```
RTC
[RTC]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 13       Routes : 13

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0          RD   172.16.103.1    GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.102.0/24  Direct  0    0           D   172.16.102.2    GigabitEthernet
0/0/1
   172.16.102.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
 172.16.102.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/1
   172.16.103.0/24  Direct  0    0           D   172.16.103.2    GigabitEthernet
0/0/0
   172.16.103.2/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.103.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.10.0/24  Static  60   0          RD   172.16.102.1    GigabitEthernet
0/0/1
   192.168.30.0/24  Static  60   0          RD   172.16.103.1    GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTC]
```

Q4-1.19. Paste the screenshot of the IP routing table on the router RTD.

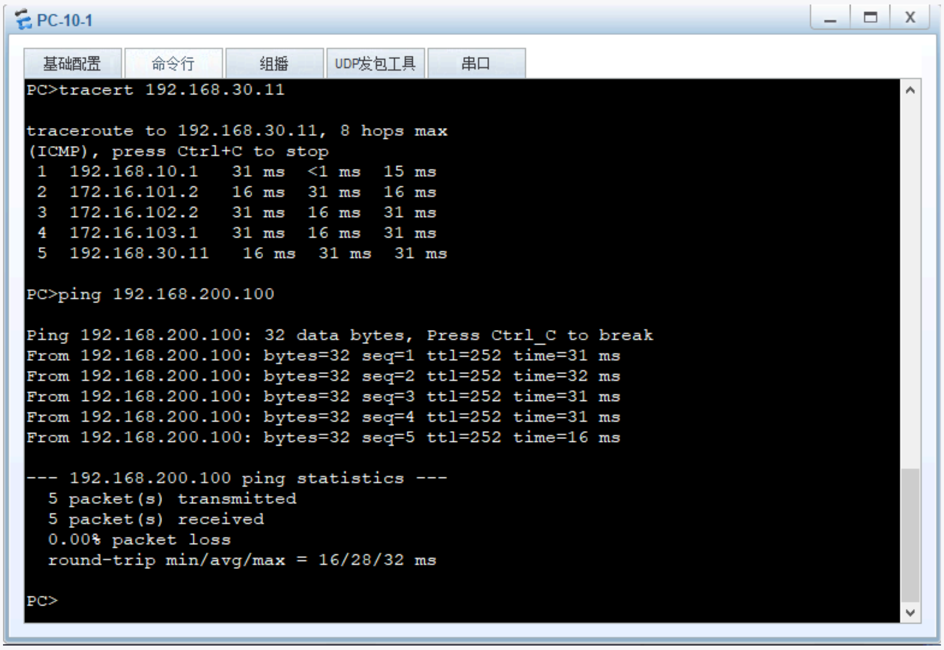


```
RTD
[RTD]
[RTD]display ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 14       Routes : 14

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   Static  60   0          RD   172.16.103.2    GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
   172.16.103.0/24  Direct  0    0           D   172.16.103.1    GigabitEthernet
0/0/0
   172.16.103.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
 172.16.103.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.30.0/24  Direct  0    0           D   192.168.30.1    GigabitEthernet
0/0/2
   192.168.30.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
 192.168.30.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/2
  192.168.200.0/24  Direct  0    0           D   192.168.200.100 LoopBack0
192.168.200.100/32  Direct  0    0           D   127.0.0.1       LoopBack0
192.168.200.255/32  Direct  0    0           D   127.0.0.1       LoopBack0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[RTD]
```

Q4-1.20. Can PC-10-1 and Loopback 0 on RTD ping each other? Please paste the screenshot of the ping result. If they can ping each other, please explain why.

可以。因为现在双向路由均完整，发送与接收线路上默认路由与静态路由协同工作，可以确保数据包正确转发。具体而言：

- **去程**：虽然没有配置到达 192.168.200.0/24 的明细路由，但由于配置了 0.0.0.0/0 的默认路由，路由器匹配不到具体网段时会将其作为"兜底"规则，将数据包一路向右转发，最终到达 RTD 并被接收。
- **回程**：RTD 删除了之前的明细路由，因此回复给 PC-10-1 的数据包同样会命中 RTD 上的默认路由 0.0.0.0/0，被一路向左转发回 PC-10-1。




```
PC>tracert 192.168.30.11

traceroute to 192.168.30.11, 8 hops max
(ICMP), press Ctrl+C to stop
 1  192.168.10.1   31 ms  <1 ms  15 ms
 2  172.16.101.2   16 ms  31 ms  16 ms
 3  172.16.102.2   31 ms  16 ms  31 ms
 4  172.16.103.1   31 ms  16 ms  31 ms
 5  192.168.30.11   16 ms  31 ms  31 ms

PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=2 ttl=252 time=32 ms
From 192.168.200.100: bytes=32 seq=3 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=4 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=5 ttl=252 time=16 ms

--- 192.168.200.100 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 16/28/32 ms

PC>
```